

Models and Formulas of the Photovoltaic
Layout and Shading Simulator
Mathematical Appendix — Traceability of Source Code Formulas

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Abstract

This appendix retraces the calculation models implemented in the Python/Tkinter photovoltaic layout and shading application. It retains the typography, mathematical notation and section structure of the original engineering note, while updating the formulas for dated solar simulations, two-pole DC routes, electrical checks and hourly PV/BESS operation. All values produced by these models are estimates subject to the stated input and validation conditions.

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1 Notations

Symbol	Unit	Meaning
ϕ, λ	°	Latitude north-positive; longitude east-positive
n, N_y	day	Day number; 365 or 366 days in calculation year
h, u	h	Local decimal clock hour; UTC offset
γ, δ	rad	Fractional year angle; declination
E_t	min	Equation of time
ω, α, A_s	°	Hour angle; solar elevation; azimuth from north clockwise
$\beta, \gamma_p, \theta_i$	°	Plane tilt; plane azimuth; incidence angle
$G_{\text{DNI}}, G_{\text{POA}}$	W/m ²	Direct normal and plane-of-array irradiance
T_c, T_{amb}	°C	Cell and ambient temperatures
P_{STC}, μ_P	W, %/°C	STC module power; power temperature coefficient
f_s, o	—	Shaded fraction; obstacle opacity, both in [0, 1]
L_A, L_B, S, ρ	m, m, mm ² , Ω mm ² /m	Positive and negative cable lengths; section; resistivity
C_{nom}, Q	kWh	Nominal battery capacity; stored energy
$E_{\text{PV}}, E_{\text{load}}$	kWh	Hourly AC production and refrigerator demand

Table 1: Main symbols; additional symbols are defined at their first occurrence.

Trigonometric relations use radians internally; displayed angles are in degrees unless specified. Electrical energy is in Wh in the shading simulation and kWh in the hourly AC balance.

2 Solar Position

Source: `mixins/shadow_energy.py: _compute_solar_position`

The position model uses a fractional-year approximation for declination and equation of time. The preview with day/month alone uses the representative year 2024. Dated shadow and annual simulations use the actual year; the annual Europe/Rome profile uses the UTC offset of each individual local hour, including daylight-saving transitions.

2.1 Fractional Year Angle

$$\gamma = \frac{2\pi}{N_y} \left(n - 1 + \frac{h - 12}{24} \right), \quad N_y \in \{365, 366\}. \quad (1)$$

2.2 Equation of Time

$$E_t = 229.18(0.000075 + 0.001868 \cos \gamma - 0.032077 \sin \gamma - 0.014615 \cos(2\gamma) - 0.040849 \sin(2\gamma)) \quad [\text{min}]. \quad (2)$$

2.3 Solar Declination

$$\begin{aligned} \delta = & 0.006918 - 0.399912 \cos \gamma + 0.070257 \sin \gamma \\ & - 0.006758 \cos(2\gamma) + 0.000907 \sin(2\gamma) \\ & - 0.002697 \cos(3\gamma) + 0.001480 \sin(3\gamma). \end{aligned} \quad (3)$$

2.4 True Solar Time and Hour Angle

$$t_{\text{offset}} = E_t + 4\lambda - 60u \quad [\text{min}], \quad (4)$$

$$t_{\text{solar}} = 60h + t_{\text{offset}} \quad [\text{min}], \quad (5)$$

$$\omega = \frac{t_{\text{solar}}}{4} - 180 \quad [^\circ]. \quad (6)$$

2.5 Geometric Solar Elevation

$$\alpha_{\text{geo}} = \arcsin\left(\text{clip}_{[-1,1]}(\sin \phi \sin \delta + \cos \phi \cos \delta \cos \omega)\right). \quad (7)$$

2.6 Atmospheric Refraction Correction

$$\Delta\alpha_{\text{ref}} = \begin{cases} \frac{1.02}{\tan\left(\alpha_{\text{geo}} + \frac{10.3}{\alpha_{\text{geo}} + 5.11}\right)} & \alpha_{\text{geo}} > -0.85^\circ, \\ 0 & \text{otherwise} \end{cases} \quad [\text{arcmin}], \quad (8)$$

$$\alpha = \alpha_{\text{geo}} + \frac{\Delta\alpha_{\text{ref}}}{60}. \quad (9)$$

The constants and angles inside the refraction tangent are expressed in degrees; the implementation converts them to radians before calling trigonometric functions.

2.7 Solar Azimuth

$$c_A = \text{clip}_{[-1,1]} \left(\frac{\sin \delta - \sin \alpha_{\text{geo}} \sin \phi}{\cos \alpha_{\text{geo}} \cos \phi} \right), \quad (10)$$

$$A_s = \begin{cases} 360^\circ - \arccos(c_A) & \omega > 0, \\ \arccos(c_A) & \omega \leq 0. \end{cases} \quad (11)$$

Model limitation. At a singular zenith/polar denominator, the implementation applies a conventional azimuth instead of evaluating the quotient. These position equations are approximations, not an ephemeris-grade solar-position algorithm.

3 Solar Irradiance on Module Plane

Source: `mixins/shadow_energy.py: get_clear_sky_poa_irradiance`

The shadow-study scenario uses a simplified clear-sky sky model: $G_{\text{sc}} = 1367 \text{ W/m}^2$, $\tau = 0.7$ and ground albedo 0.20. Imported historical weather, when supplied for the annual energy simulation, replaces the clear-sky plane irradiance and supplies hourly ambient temperature for one orientation.

3.1 Direct Normal Irradiance

$$m_a = \frac{1}{\sin \alpha}, \quad G_{\text{DNI}} = \begin{cases} G_{\text{sc}} \tau^{m_a^{0.675}} & \alpha > 0.5^\circ \text{ and } \sin \alpha \geq 0.01, \\ 0 & \text{otherwise.} \end{cases} \quad (12)$$

3.2 Angle of Incidence

$$c_i = \text{clip}_{[0,1]} (\sin \alpha \cos \beta + \cos \alpha \sin \beta \cos(A_s - \gamma_p)). \quad (13)$$

3.3 Direct, Diffuse and Reflected Components

$$G_{\text{dir}} = G_{\text{DNI}} c_i, \quad (14)$$

$$G_{\text{h,diff}} = 0.12 G_{\text{DNI}} \max(0, \sin \alpha), \quad (15)$$

$$G_{\text{diff}} = G_{\text{h,diff}} \frac{1 + \cos \beta}{2}, \quad (16)$$

$$G_{\text{h}} = G_{\text{DNI}} \max(0, \sin \alpha) + G_{\text{h,diff}}, \quad (17)$$

$$G_{\text{alb}} = 0.20 G_{\text{h}} \frac{1 - \cos \beta}{2}. \quad (18)$$

3.4 Total Plane Irradiance

$$G_{\text{POA}} = \max(0, G_{\text{dir}} + G_{\text{diff}} + G_{\text{alb}}). \quad (19)$$

Model limitation. The diffuse share is a fixed empirical fraction, and albedo is constant. The clear-sky scenario is not a real-weather production guarantee. Imported weather is historical input and must represent the selected plane orientation.

4 Module Electrical Model

Source: `mixins/shadow_energy.py: estimate_panel_power_w`

4.1 Effective Irradiance and Cell Temperature

$$G_{\text{eff}} = G_{\text{POA}}(1 - f_s), \quad T_c = T_{\text{amb}} + (T_{\text{NOCT}} - 20) \frac{G_{\text{eff}}}{800}. \quad (20)$$

$T_{\text{amb}} = 20^\circ\text{C}$ is assumed only for the clear-sky shadow study; imported hourly weather supplies T_{amb} where enabled.

4.2 Temperature Factor

$$k_T = \max\left(0, 1 + \frac{\mu_P}{100}(T_c - 25)\right). \quad (21)$$

4.3 Instantaneous Power

$$P(t) = \begin{cases} P_{\text{STC}} \frac{G_{\text{eff}}}{1000} k_T & G_{\text{eff}} > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (22)$$

4.4 Energy over One Step

For midpoint $t_m = (t_0 + t_1)/2$ and Δt in hours:

$$E_{\text{ideal}} = P_{f_s=0}(t_m) \Delta t, \quad (23)$$

$$E_{\text{real}} = P_{f_s}(t_m) \Delta t, \quad (24)$$

$$E_{\text{loss}} = \max(0, E_{\text{ideal}} - E_{\text{real}}). \quad (25)$$

Model limitation. This is a linear irradiance and temperature power estimate. It does not solve a diode I-V equation, diode bypass behaviour or series mismatch.

5 Time Integration of Shading Simulation

Source: `mixins/shadow_simulation.py: run, data_at`

$$E_{\text{total}} = \sum_{k=1}^M P(t_{m,k}) \Delta t_k, \quad t_{m,k} = \frac{t_{0,k} + t_{1,k}}{2}. \quad (26)$$

The time step controls the accuracy near sunrise, sunset and abrupt shadow changes.

5.1 Aggregation per String

Source: `mixins/shadow_energy.py: summarize_string_group`

For the M_s modules of a string:

$$E_{\text{ideal,string}} = \sum_{j=1}^{M_s} E_{\text{ideal,j}}, \quad E_{\text{loss,string}} = \sum_{j=1}^{M_s} E_{\text{loss,j}}, \quad (27)$$

$$r_{\text{loss,string}} = 100 \frac{E_{\text{loss,string}}}{E_{\text{ideal,string}}}, \quad (28)$$

$$\Delta_{\text{spread}} = \max_j(r_{\text{loss,j}}) - \min_j(r_{\text{loss,j}}). \quad (29)$$

The last quantity is a shading heterogeneity indicator, not a calculated series-string electrical mismatch loss.

6 Cast Shadow Geometry

Source: `mixins/shadow_geometry.py: compute_shadow_geometry`

The simplified obstacle is a vertical mast with a square plan footprint, projected separately onto each roof-zone elevation.

6.1 Polygon Area

$$A = \frac{1}{2} \left| \sum_{k=1}^p (x_k y_{k+1} - x_{k+1} y_k) \right|. \quad (30)$$

6.2 Projection Distance

$$d_{\text{shadow}} = \frac{H_{\text{mast}} - H_{\text{roof}}}{\tan \alpha}, \quad H_{\text{mast}} > H_{\text{roof}}, \quad \alpha > 0. \quad (31)$$

The footprint is translated opposite to the sun azimuth, accounting for the user-defined image north rotation. The convex hull of original and translated footprint vertices produces the candidate shadow polygon.

6.3 Convex Hull and Polygon Intersection

The convex hull is computed with Andrew’s monotone-chain procedure, with complexity $O(p \log p)$. Intersection with the module rectangle uses successive Sutherland–Hodgman clipping against each rectangle edge.

6.4 Shaded Module Fraction

$$f_s = \text{clip}_{[0,1]} \left(o \frac{A_{\text{intersection}}}{A_{\text{module}}} \right), \quad s\% = 100f_s. \quad (32)$$

Model limitation. A zero or negative height difference, or a sun below the horizon, produces no modeled projected shadow. The obstacle geometry and optical transparency remain simplified.

7 Panel Grid Layout

Source: `mixins/zone_tools.py: recalculate_zone_grids, zone_rotate_point`

7.1 Rows and Columns

$$n_{\text{col}} = \left\lfloor \frac{W}{w} \right\rfloor, \quad n_{\text{row}} = \left\lfloor \frac{H}{h} \right\rfloor. \quad (33)$$

These dimensions describe the preliminary rectangular grid before individual edits, roof setbacks and actual mounting constraints.

7.2 Alignment

With $r_x = W - n_{\text{col}}w$ and $r_y = H - n_{\text{row}}h$:

$$o_x \in \{0, r_x/2, r_x\}, \quad o_y \in \{0, r_y/2, r_y\}. \quad (34)$$

7.3 Rotation about Each Module Centre

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} c_x \\ c_y \end{pmatrix} + \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x - c_x \\ y - c_y \end{pmatrix}. \quad (35)$$

8 DC Wiring

Source: `mixins/two_pole_cables.py: get_two_pole_route`

8.1 Two Physical Routes

For endpoints $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$, a direct orthogonal fallback has length

$$L_{\text{orth}} = |x_2 - x_1| + |y_2 - y_1|. \quad (36)$$

The drawn pathway network instead uses shortest paths through a graph (Dijkstra). The string's first and last terminals each receive a separate route; modeled vertical roof-to-inverter drops and end reserves are included:

$$L_A = L_{\text{roof},A} + |H_A - H_{\text{inv}}| + R_A, \quad (37)$$

$$L_B = L_{\text{roof},B} + |H_B - H_{\text{inv}}| + R_B, \quad L_{\text{loop}} = L_A + L_B. \quad (38)$$

Model limitation. Routes and risers are preliminary estimates until actual cable trays, inverter elevation, termination points and inter-module wiring have been surveyed. Recalculate routes after editing panel positions or pathways.

8.2 Voltage Drop and Selected Cross-Section

Source: `mixins/notes_tools.py`: `calculate_dc_cable_size`

$$\Delta U_{\text{DC}} = \frac{I\rho(L_A + L_B)}{S}, \quad S_{\text{min}} = \frac{I\rho(L_A + L_B)}{U\varepsilon_{\text{target}}/100}, \quad (39)$$

$$S_{\text{norm}} = \min\{s \in \mathcal{S}_{\text{norm}} : s \geq S_{\text{min}}\}, \quad \varepsilon_{\text{actual}} = 100 \frac{I\rho(L_A + L_B)}{S_{\text{norm}}U}. \quad (40)$$

Here $\mathcal{S}_{\text{norm}} = \{1.5, 2.5, 4, 6, 10, 16, 25, 35, 50, 70, 95, 120, 150, 185, 240\} \text{ mm}^2$. In the general-purpose calculator, a specified one-way length L represents $L_A = L_B = L$, giving $L_{\text{loop}} = 2L$. The project default resistivity is $0.017 \Omega \text{ mm}^2/\text{m}$ and is user-editable.

Model limitation. Voltage drop alone cannot determine conductor acceptability. Installation temperature, grouping, ampacity, insulation, DC short-circuit capability, earthing and protective-device coordination require separate engineering checks.

9 Electrical Summary under STC Conditions

Source: `project_validation.py`; `detailed_electrical.py`

9.1 One String of N Modules in Series

$$P_{\text{dc},s} = \frac{NP_{\text{STC}}}{1000} [\text{kWp}], \quad V_{\text{oc},s} = NV_{\text{oc}}, \quad V_{\text{mpp},s} = NV_{\text{mpp}}, \quad (41)$$

$$I_{\text{sc},s} = I_{\text{sc}}, \quad I_{\text{mpp},s} = I_{\text{mpp}}. \quad (42)$$

9.2 Parallel Strings on One MPPT

$$P_{\text{dc},m} = \sum_{s \in m} P_{\text{dc},s}, \quad I_{\text{sc},m} = \sum_{s \in m} I_{\text{sc},s}, \quad I_{\text{mpp},m} = \sum_{s \in m} I_{\text{mpp},s}. \quad (43)$$

Strings of unequal series lengths on the same MPPT are flagged for review. An interval of string voltages does not demonstrate their electrical compatibility; module technology, operating temperature, currents and manufacturer input limits must be verified.

9.3 Cold Open-Circuit Voltage and Inverter Totals

$$V_{\text{oc},s}(T_{\text{min}}) = NV_{\text{oc,STC}} \left[1 + \frac{\mu V_{\text{oc}}}{100} (T_{\text{min}} - 25) \right], \quad (44)$$

$$P_{\text{dc,inv}} = \sum_{s \text{ assigned}} P_{\text{dc},s}, \quad I_{\text{ac,nominal}} = \frac{1000 S_{\text{ac,kVA}}}{\sqrt{3} U_{\text{LL}}}. \quad (45)$$

The cold-voltage calculation needs the module coefficient and design minimum cell temperature. The AC expression assumes three-phase operation and a specified line-to-line voltage; it is not a breaker setting. The project arrangement uses three hybrid inverters of nominal 125 kW with 10 MPPT each. Product-specific operating windows and BESS interfaces must be confirmed from the exact equipment documentation.

10 Hourly PV, Refrigerator Load and BESS

Source: `self_consumption.py`; `battery_dispatch.py`

The following AC-side accounting is performed independently for each local hour ($\Delta t = 1$ h). For the PV-only option, set charge and discharge to zero.

10.1 Direct Use and Storage Dispatch

$$E_{\text{direct}} = \min(E_{\text{PV}}, E_{\text{load}}), \quad (46)$$

$$E_{\text{surplus}} = E_{\text{PV}} - E_{\text{direct}}, \quad E_{\text{deficit}} = E_{\text{load}} - E_{\text{direct}}. \quad (47)$$

Let $Q \in [Q_{\min}, Q_{\max}]$ be stored energy, and let $\eta_c = \eta_d = \sqrt{\eta_{\text{roundtrip}}}$. At each hour:

$$E_{\text{charge}} = \min\left(E_{\text{surplus}}, P_c \Delta t, \frac{Q_{\max} - Q}{\eta_c}\right), \quad (48)$$

$$Q' = Q + \eta_c E_{\text{charge}}, \quad (49)$$

$$E_{\text{discharge}} = \min(E_{\text{deficit}}, P_d \Delta t, (Q' - Q_{\min})\eta_d), \quad (50)$$

$$Q_{\text{next}} = Q' - \frac{E_{\text{discharge}}}{\eta_d}. \quad (51)$$

The default battery is provisionally identified as *Deye MC-L522-2H3*, with nominal $C_{\text{nom}} = 522.496$ kWh per cabinet, $Q_{\min} = 0.10C_{\text{nom}}$, $Q_{\max} = 0.90C_{\text{nom}}$, initial $Q = Q_{\min}$ and $\eta_{\text{roundtrip}} = 0.90$. The configured limits $P_c = 250$ kW and $P_d = 150$ kW are *shared plant limits*: two cabinets double energy capacity but do not double power. Charging from the grid and resale of discharged battery energy are not modeled.

10.2 Grid Import, Export and Curtailment

$$E_{\text{grid}} = E_{\text{deficit}} - E_{\text{discharge}}, \quad (52)$$

$$E_{\text{export}} = \min(E_{\text{surplus}} - E_{\text{charge}}, P_{\text{export}} \Delta t), \quad (53)$$

$$E_{\text{curtailed}} = E_{\text{surplus}} - E_{\text{charge}} - E_{\text{export}}, \quad (54)$$

$$E_{\text{useful self}} = E_{\text{direct}} + E_{\text{discharge}}, \quad (55)$$

$$E_{\text{PV retained}} = E_{\text{direct}} + E_{\text{charge}}. \quad (56)$$

The last two totals differ because of battery losses and changes in stored energy; they must not be added. Checks on annual aggregates are $E_{\text{load}} = E_{\text{useful self}} + E_{\text{grid}}$ and $E_{\text{PV}} = E_{\text{PV retained}} + E_{\text{export}} + E_{\text{curtailed}}$. The existing grid contract is recorded as 507 kW, and 300 kW export is the *requested, not approved* supply limit.

11 Annual Economics and Scenario Comparison

Source: `battery_dispatch.py`; `mixins/self_consumption_ui.py`

Annual energy quantities are sums of the hourly quantities above. Let p_b be the purchase price in EUR/kWh and p_s the credit for exported energy in EUR/kWh. Then for each scenario j :

$$C_0 = p_b E_{\text{load, annual}}, \quad (57)$$

$$C_j = p_b E_{\text{grid}, j} - p_s E_{\text{export}, j}, \quad (58)$$

$$B_j = C_0 - C_j, \quad (59)$$

$$T_j = \begin{cases} K_j / B_j & B_j > 0, \\ \text{undefined} & B_j \leq 0. \end{cases} \quad (60)$$

C_j is the *net annual electricity cost* after export credit; B_j is the annual benefit relative to buying the entire refrigerator demand. For PV only, one battery and two batteries, illustrative upfront capital K_j is respectively EUR 250,000, 300,000 and 350,000. Evaluate $p_b = 0.25$ and 0.32 EUR/kWh and a specified export tariff in the proposed 0.07–0.08 EUR/kWh range. A baseline of EUR 125,000/year at EUR 0.25/kWh corresponds to 500,000 kWh/year *only if* the metered annual demand confirms that figure. The simple payback T_j excludes operating expenses, tax, financing, degradation, battery replacement and future tariff changes.

12 Summary of Model Limitations and Required Engineering Checks

- The default sky is simplified clear sky. Hourly historical weather can be imported, but prediction and uncertainty still depend on input quality.
- Power scaling and string shading do not simulate diode bypass or I–V curves. The spread indicator is diagnostic only.
- Inverter count, MPPT allocation, cold V_{oc} , hot V_{mpp} , parallel current and any bifacial contribution need actual datasheets.
- Two-pole DC routes include modeled vertical drops and reserves; field routing, AC cables, inter-module leads and protection studies remain to be completed by the responsible designer.
- BESS power and efficiency are configurable assumptions; the listed model and interface must be confirmed before procurement. Export at 300 kW remains subject to grid-operator approval.